Dragon Boat Canada
Stroke Fundamentals and Drills

Coaching Manual Level 1

Source: Dragon Boat Canada, Section 12.4 Stroke Fundamentals and Module 13 Drills. © Dragon Boat Canada.

Factors affecting skill execution

- Environment: Rough conditions, boat riding high or low, head/tail wind, temperature.
- Mental/Cognitive/Emotional: Are there cognitive or emotional obstacles to the athlete's execution of the skill - lack of understanding, confusion, lack of concentration, fear, motivation, or anxiety?
- Physical/Motor: What physical factors are affecting the athlete's performance of the skill - Strength, Balance, Flexibility, Fitness?

Coaches should attempt to correct other factors first before attempting to correct physical ones. These corrections are easier to implement than physical instruction and will cause persistent deficiencies if not addressed. For instance, if an athlete has a paddle that is too long for them to properly execute their recovery, no physical instruction or drill will be able to fix their stroke.

Module 13 Drills

This will provide a few tried and tested drills to help a coach help their paddlers. It is always good practice to keep notes for oneself on what was effective and what can be modified for the next time. Collaborate with the coaches around you. Ask for their ideas and suggestions on how to help paddlers improve their stroke, and/or help the team work as a unit.

Good luck. Helping paddlers start with good technique will be beneficial to everyone.

Stroke Fundamentals

Stroke phase	Basic Position/Motion	Advanced Load	High Performance Speed
Set-up loop	• Core/Torso posture • Stable seated position • Hip-to-shoulder rotation at setup	• Strong, open extended top arm • Bottom-arm extension • Lower-body load - loaded leg and rotated hips • Tall body - hinge to use for entry	• Effective reach vs. absolute reach • Complete set-up before moving to water
Catch	• Move body to water • Paddle 90° or positive at bury • Held rotation • Fully buried blade before pull	• Applying weight to the water • Finding support from the water • Dropping bottom shoulder/lat to catch • Loading top arm through body • Finding load through body, not with arms	• Quick engagement to support • Clean entry • Establishing load at entry
Pull	• Lower body as momentum generator • Hip - Core - Shoulder sequence of motion • Arms follow body through stroke • Constant depth of pull • Movements continue throughout stroke	• Lift (sit-up) and rotate at the same time • Lower (paddle-side) lat engaging • Lift torso through lats from load on the water • Downward pressure on blade • Top arm tracks down and back, not forward	• Constant load on paddle, acceleration through the stroke • Body ahead of hands/paddle through stroke
Exit/Recovery	• Body always slightly forward; athlete tall at exit • Balanced (stable) movement through recovery • Top arm not dropping below sternum • Exit position at or before hip (high variability between crews)	• Continued paddle pressure to exit • Using load of water to begin moving body forward • Reload body through recovery • Body moving forward to next set-up	• Quick disengagement from water - minimum time from load to paddle out of water • Controlled recovery at speed - complete reload quickly to set-up • Quiet recovery - minimum extraneous motion during recovery

Drills

Name	Focus	Drill
Push and Pull	Find the heavy water. Use the core for rotation.	During this drill, the paddle never comes out of the water. • Have paddlers start with a fully buried paddle up front. • Paddlers use their best technique to pull the paddle through the stroke. • At the exit, paddlers move the paddle back to the front of the stroke, staying in the water. • The next stroke should feel heavier as the boat should be moving slightly backwards. • Do this several times so paddlers can feel what a connected stroke feels like.
7-Up	For boat synchronicity: the goal is to have everyone move together, no rushing, no lagging.	• The whole boat takes 1 stroke together, returns to the front of the stroke, and lets the boat glide for a second or two. • The whole boat takes 2 continuous strokes together and waits at the front together for a few seconds. • The whole boat takes 3, then 4, then 5, then 6, then 7 strokes together.
Sectional Paddling	To have paddlers look down the boat to see what is happening up front, and stay in unison.	There are many ways to do sectional paddling; below are a few examples. • Fronts, Middles, Back. • Right Front Half and Left Back Half, followed by Left Front Half and Right Back Half. • Only odd-numbered seats, followed by only even-numbered seats. • Left-side odd-numbers and right-side even-numbers, followed by right-side odd-numbers and left-side even-numbers.
Frankenstein	Focuses on core rotation in order to make the paddle move through the water.	• Paddlers are told they cannot bend their elbows throughout the stroke. The only way they can make the paddle move is by using core rotation and cycling the legs and hips.
Hand on Forehead	Focuses on the paddler keeping the top hand away from the body throughout the stroke.	• The paddler puts their top hand on their forehead and keeps it there throughout the stroke, which forces an over-bent top arm. • The goal is to have the paddler practise this ineffective way of paddling.
Legs Cycling	Focuses on having paddlers use their legs and hips to generate power.	• Paddlers put the paddle-side foot forward and the other behind. • Paddlers put the non-paddle-side foot forward and the paddle-side leg behind. • Paddlers put both legs behind. • Paddlers put both legs forward. • After each leg position, have paddlers share how they felt their paddling changed.
Tall Paddling for Breathing	To show paddlers how body position affects breathing ability.	• Have paddlers crunch themselves into a tight ball and take 5 deep breaths. • Have paddlers sit tall on their seat and take 5 deep breaths. • Ask paddlers to share how they felt their body position affected their ability to breathe.

Name	Focus	Drill
Don't Spill the Champagne (or beer)	Focus the paddler on keeping the paddle arm straight and parallel to the water on the extension.	Tell paddlers that they are holding a fine glass of champagne in their bottom hand. • Have them move through the stroke, focusing on keeping their hand and bottom arm parallel to the surface of the water, i.e. not spilling the champagne.
Upside Down Paddle	Allows paddlers to see that good technique will move the dragon boat even using the handle.	• Have all paddlers turn their paddle upside down. • Have paddlers move through the stroke with their paddle upside down. • Let paddlers know they need to find the heavy water and not just push the paddle through.
Your Paddle Talks to You	Have paddlers use all senses to improve their paddling.	Have paddlers listen to the sounds the paddle makes when paddling. • A solid catch should be quiet with limited splash. • A solid exit should have limited splash and also be quiet.
Hand on Your Back	To have paddlers feel when they are over-rotating.	• Have even-numbered paddlers hold their hand at the back of the paddler ahead of them at the 90° point (upright). • Have paddlers go through several strokes. Their back should not touch the hand of the person behind them. • Paddlers should use this kinesthetic feedback to adjust their stroke.
Paddling "Blind"	Encourages paddlers to "feel" the boat.	• Have all paddlers paddle through 10 strokes to get the boat moving. • On the call, have all paddlers close their eyes and encourage them to feel the movement of the boat. • Have paddlers start with a slower stroke than normal so they have time to adjust to paddling without seeing. • Paddlers can increase the rate as they are more able to paddle using the motion of the boat rather than vision.
Find your Entry Point	To have paddlers understand their optimal hang and entry point.	• Have paddlers put their paddle in the water at their exit point - FULLY BURIED. • Maintaining a fully buried paddle, have paddlers move their blade through the water toward the catch point. The blade must remain fully buried. • Paddlers should mark on the gunnel, using wax or chalk, the point at which they could reach while fully buried. • This should become the point where they reach in the air so they can drop to a fully buried paddle immediately.
Hang Time - Float it Back	Focus on not rushing back (air time), using the boat's momentum, and feeling the glide.	• Use a 2:1 or 3:1 ratio. Have paddlers use their regular paddle stroke in the water and then float it back toward the front. • By varying the ratio, paddlers have to focus on keeping together as a team and see how the boat will carry the momentum forward. • Distance can be measured for 20 strokes using minimal air time (1:1), and compared to 20 strokes with more air time (1:3).

Name	Focus	Drill
¼, ½, ¾, whole paddle	Focus on seeing a fully buried paddle versus a partially buried angled paddle.	• Have paddlers hold the paddle with just the corner tip (¼ paddle) in the water and paddle. • Have paddlers hold the paddle with the angled paddle (½ paddle) in the water and paddle. • Have paddlers hold the paddle with the angled paddle (¾ paddle) in the water and paddle. • Have paddlers hold the whole paddle in the water and paddle.
Use Your Body	Focus on how to use the entire body to move their paddle.	Take time between each set of 10 strokes to explain and demonstrate how the addition of each body part should happen correctly. • First, paddlers take 10 strokes using only their hands to move the paddle. • Second, paddlers take 10 strokes using hands and arms. • Next 10 strokes add in the use of shoulders. • Next 10 strokes add in core muscles with rotation. • Next 10 strokes add in hip rotation. • Next 10 strokes add hip cycling by pushing off the heels. • By now the entire body should be moving and involved with the stroke. Hopefully paddlers feel the added power of their stroke with the addition of each body part.
Starts	First Strokes: to propel the boat from a dead stop to moving in a powerful manner.	• Usually 3-5 strokes. • Should be slow, fully buried blade strokes. • Should be ½-¾ length of a full-extension stroke. • Envision lifting the boat up from a dead position. • The first stroke may start in the water, over the water, at the tips, or half buried, depending on the situation. • The body should be leaning forward and to the outside. • The body should be “itching” to go. • First motion is down. • Remind paddlers not to lurch forward or fall back on the recovery (a strong core is important).
Starts continued	Middle Acceleration Strokes	• Usually 4-15 strokes. • Should be fast. • Full-blade strokes with ½-¾ length of full extension. • Almost too fast, but needs to be powerful. • The number of strokes will vary based on the strength, weight, and skill of the team.
Starts continued	Transition	• Usually 3-5 strokes. • Once top speed is reached, the team needs to transition to the racing stroke. • Transitioning to the full stroke should happen over the 3-5 strokes, and not be done in just 1. • Who calls for the transition varies, but often it is the drummer or the steer. • The goal is to find the longest and most powerful stroke the team can manage for the length of the race. • With new crews, it is often useful to have the drummer or steer count the strokes to help keep everyone in time.

Name	Focus	Drill
Part Paddling	Focus on Timing	Build up the boat speed, then let it run. While the boat is gliding on the water, instruct: • Two paddlers at a time take ten strokes each with their seat partner only. Start with seat one, proceed through to seat ten, and repeat back up the boat. • Do this same drill with 4 paddlers at a time (i.e. first 4, next 4, back 6). • Repeat with the front half of the boat and the back of the boat. • Repeat with the odds and evens.
Pause Strokes	Focus on Timing	• Have paddlers put their paddles up just above the water, prepared to take a stroke. Give the command to take one stroke and put paddles back up in the starting position. Repeat several times, then increase to two strokes, three strokes, etc., up to ten strokes. • Holding the paddle in the start position above the water is a strain on the shoulders. Be sure to allow adequate rest between sets of strokes.
Stroke Rate	Focus on Timing	• Have paddlers paddle at a slow rate for ten strokes, then count “3-2-1-Up!” to have them increase the rate for ten strokes. • Repeat several times.
Paddle Position - Angled Entries	Focus on Correct Paddle Position	• If the bottom hand reaches farther than the top hand and the blade spears the water instead of entering vertically, straighten the top arm or check for over-rotation. • If the top hand reaches farther from the body, the blade may enter at a negative angle and act as a shovel. Extend the bottom hand so both hands reach equally and the paddle enters almost vertically with only a slight spearing angle.
Paddle Position - Rotated Blade	Focus on Correct Paddle Position	• If the blade faces the gunwale or away from the boat, review the grip. • The knuckles and thumbs of both hands should face straight ahead toward the front of the boat.
Paddle Position - Paddle Swings Away From Side of Boat	Focus on Correct Paddle Position	• Most likely this is caused when the paddler swings or drops their top hand, causing the blade portion of the paddle to swing out and away from the boat. • Keep the top hand up, in front of their face, and over the side of the boat. • Focus on keeping the thumb of the bottom hand close to the side of the boat.
Rotation	Using Only Arms	• The paddler is reaching forward with only their arms. • Work on rotating the body from the waist, turning the core of the body from the hips. Model this action on land and seated in the boat without using a paddle. • Have the paddler reach with their arms. Hold their paddle in the middle of the shaft and pull it forward slightly and gently to help them feel the rotation from their hips. • Once rotation has been mastered, work on bending down towards the water at the waist.

Name	Focus	Drill
Rotation	Lunging	• The paddler reaches with their arms and then lunges their body weight down to the water without rotating the hips or waist. • This can affect the boat run through improper weight distribution. • Focus on core strength, keeping the outside arm parallel to the water/gunnel.
Rotation	Over Rotation	• This generally causes problems with paddle position. It is noticeable by a bent top-arm position or when the top arm is behind the head. • The arms tend to follow the body if they are not straight. The paddle may hit the side of the boat just before the paddle enters the water. • Check core body position. • Check rotation at the hips.
Catch and Pull	Excessive Splashing at the Catch	• Straighten out the top arm and instruct paddlers to move top and bottom hands together. • The paddle is likely being pulled back before the full blade is buried in the water, creating a splash back. • The paddler must bury the blade before beginning to pull back. • For a drill, have the paddler set up, catch, and count one out loud before sitting up and pulling. This is hard to do in a moving boat, so it is best to stop the boat each time.
Catch and Pull	Shoveling at the Exit	• This is related to paddle angle. It is caused when the paddle is at a negative angle and the top hand is too far in front of the bottom hand. The bottom hand may be pulling harder than the top hand is pushing down. • The drill should focus on pressing down throughout the stroke. • Have paddlers focus on lifting the chest up to the sky as they press out the exit.